1 Vectors in Two and Three Dimensions

Question: What information specifies a line?

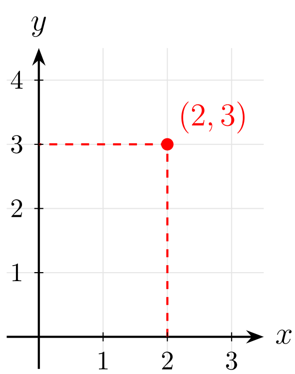
- 2 points
- A point and a direction vector
- Slope and y-intercept
- Intersection of 2 non-parallel planes

Comparison:

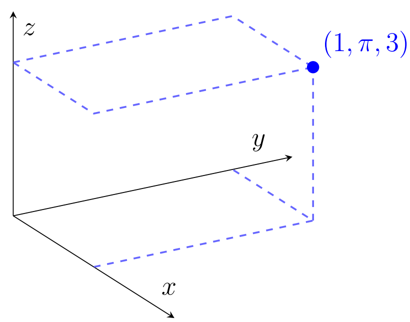
- **Single-variable calculus:** $f(x) \to$ rate of change of x
- **Multivariable calculus:** $f(x, y) \to$ functions of multiple variables

1.1 Coordinates

2-dimension (2D): $(2, 3)$



3-dimension (3D): $(1, \pi, 3)$



1.2 Vectors and Position Vectors

Given points $P = (1, 3)$, $Q = (5, 4)$ and $P' = (1, 1)$, $Q' = (5, 2)$:

> **Definition 1.1: Equivalent Vectors**
>
> Two vectors are **equivalent** if one is a translation (pure shift without rotation) of the other.

Example 1.1. Let $R = (4, 1)$ and $O = (0, 0)$. Then:

$$\overrightarrow{PQ} = \overrightarrow{P'Q'} = \overrightarrow{OR}$$

Definition 1.2: Position Vector

A vector whose initial point is at the origin $O(0, 0)$ is called a **position vector**. It is written using angle brackets:

$$\overrightarrow{OR} = \langle 4, 1 \rangle$$

Definition 1.3: Magnitude / Norm

The **length** or **magnitude** of a vector $\vec{v}$, denoted by $\|\vec{v}\|$, is the length of the line segment between its endpoints.

If $\vec{v} = \langle v_1, v_2 \rangle$, then its magnitude is given by:

$$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2}$$

For example, using $\overrightarrow{PQ} = \langle 4, 1 \rangle$:

$$\left\|\overrightarrow{PQ}\right\| = \sqrt{4^2 + 1^2} = \sqrt{17}$$

1.3 Unit Vectors

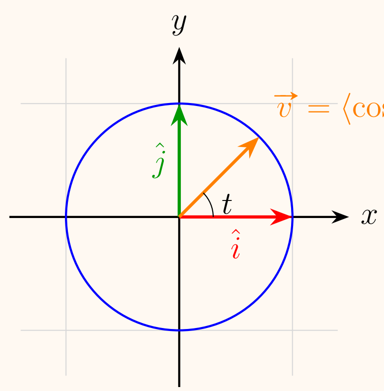
A vector $\vec{v}$ is called a **unit vector** if its magnitude is 1, i.e., $\|\vec{v}\| = 1$.

Problem 1.1: Describing Unit Vectors

Describe all unit vectors in $\mathbb{R}^2$.

A vector $\vec{v} = \langle x, y \rangle$ is a unit vector if and only if $x^2 + y^2 = 1$. Equivalently, using polar parameterization:

$$\vec{v}(t) = \langle \cos t, \sin t \rangle \quad \text{for } 0 \leq t < 2\pi$$



Standard Basis Vectors:

- **2D Basis Vectors:** $\hat{i} = \langle 1, 0 \rangle, \quad \hat{j} = \langle 0, 1 \rangle$
- **3D Basis Vectors:** $\hat{i} = \langle 1, 0, 0 \rangle, \quad \hat{j} = \langle 0, 1, 0 \rangle, \quad \hat{k} = \langle 0, 0, 1 \rangle$

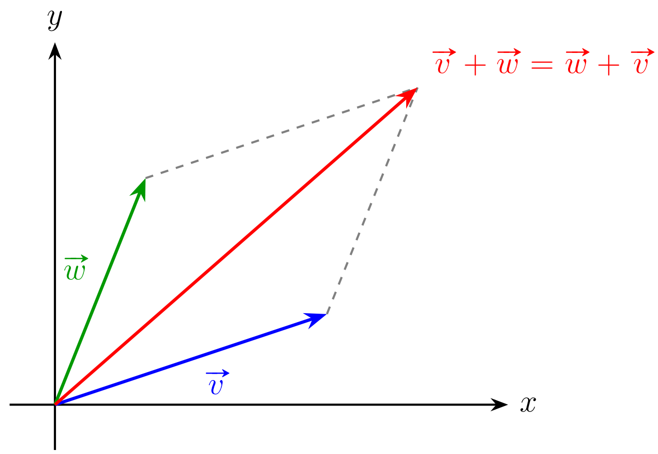
1.4 Vector Operations

Addition and Subtraction:

$$\langle v_1, v_2 \rangle + \langle w_1, w_2 \rangle = \langle v_1 + w_1, v_2 + w_2 \rangle$$
$$\langle v_1, v_2 \rangle - \langle w_1, w_2 \rangle = \langle v_1 - w_1, v_2 - w_2 \rangle \quad \text{(Equivalent to } \vec{v} + (-1)\vec{w}\text{)}$$

Scalar Multiplication: For any scalar $\lambda \in \mathbb{R}$ and vector $\vec{v} = \langle v_1, v_2 \rangle$:

$$\lambda \vec{v} = \langle \lambda v_1, \lambda v_2 \rangle$$



Important Note

Vectors are **not** designed to be divided! (Vector division is undefined).

1.5 Parallel Vectors

Definition 1.4: Parallel Vectors

Two non-zero vectors $\vec{v}$ and $\vec{w}$ ($\vec{v}, \vec{w} \neq \langle 0, 0\rangle$) are *parallel* (denoted as $\vec{v} \parallel \vec{w}$) if there exists a non-zero scalar $\lambda \in \mathbb{R}$ such that:

$$\vec{w} = \lambda \vec{v}$$

Equivalent Geometric Definition: $\vec{v} \parallel \vec{w}$ if and only if both vectors define or lie on the *same line passing through the origin.*

2 Discussion Week 1: Distance Formulae and Geometric Shapes

Definition 2.1: Distance Formula in $\mathbb{R}^2$

Let $(x, y) \in \mathbb{R}^2$ and points $P = (x_1, y_1)$, $Q = (x_2, y_2)$. The distance between P and Q is given by:

$$d(P, Q) = \left\| \vec{PQ} \right\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

This distance formula is a direct application of the Pythagorean theorem, where the distance $d(P, Q)$ forms the hypotenuse of a right-angled triangle with legs $|x_2 - x_1|$ and $|y_2 - y_1|$.

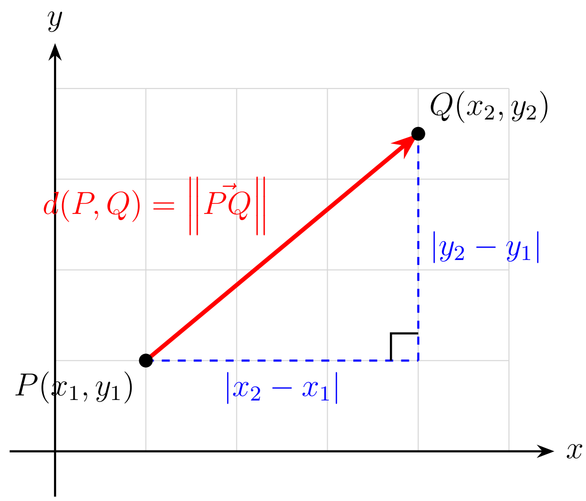


Figure 1: Vector $\vec{PQ}$ in $\mathbb{R}^2$ visualized using the Pythagorean theorem.

The case in $\mathbb{R}^3$ follows similarly:

Definition 2.2: Distance Formula in $\mathbb{R}^3$

Let $(x, y, z) \in \mathbb{R}^3$ and points $P = (x_1, y_1, z_1)$, $Q = (x_2, y_2, z_2)$. The distance between P and Q is given by:

$$d(P, Q) = \left\| \vec{PQ} \right\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

2.1 Equations of Geometric Shapes

Consider the equation $x^2 + y^2 = 4$.

In $\mathbb{R}^2$, this equation is equivalent to the set:

$$\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 4\}$$

This represents a circle with radius $r = 2$ centered at the origin. The distance from the origin $(0, 0)$ to any point (x, y) on this circle is constant and equal to 2, which can be verified directly using the distance formula:

$$\sqrt{(x-0)^2+(y-0)^2}=2$$

Hence, the geometry of the circle is directly implied by the Pythagorean theorem.

Now consider the equation $x^2+y^2=1$ in three-dimensional space, $\mathbb{R}^3$. This set is formally expressed as:

$$\{(x,y,z)\in\mathbb{R}^3 : x^2+y^2=1\}$$

Because z is completely unrestricted, the cross-section at any constant height $z=c$ forms the unit circle $x^2+y^2=1$. Consequently, this equation describes a *right circular cylinder* extending infinitely along the z-axis.

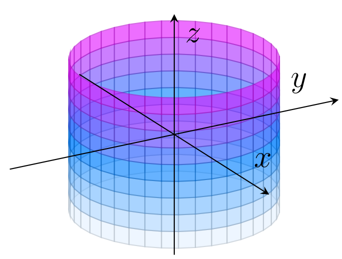


Figure 2: The equation $x^2+y^2=1$ plotted in $\mathbb{R}^3$ forms a right circular cylinder.

2.2 Linear Equations across Dimensions

In $\mathbb{R}^2$, the general form of a linear equation is given by:

$$Ax+By+C=0 \quad (A,B,C\in\mathbb{R},\ \text{with } A \text{ and } B \text{ not both zero})$$

This equation represents a *line* in two-dimensional space.

In $\mathbb{R}^3$, however, the exact same algebraic equation $Ax+By+C=0$ represents a *plane* parallel to the z-axis (since z is a free variable), not a line.

To uniquely define a line in three-dimensional space, a single linear equation is insufficient; we require a system of *two* independent linear equations (representing the intersection of two non-parallel planes).